

# Effect of temperature, precipitation and humidity on transmission of seasonal and zoonotic influenza in humans in Africa: a three-decade systematic review (1995–2025)

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## Citation

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## REVIEW TITLE AND BASIC DETAILS

### Review title

Effect of temperature, precipitation and humidity on transmission of seasonal and zoonotic influenza in humans in Africa: a three-decade systematic review (1995–2025)

### Condition or domain being studied

*Influenza; Influenza Due To Influenza A Virus; Influenza B Virus; Influenza Caused By Influenza A Virus Subtype H5; Influenza Due To Influenza A Virus Subtype H7; Influenza Due To Influenza A Virus Subtype H9*

### Rationale for the review

Influenza remains a major public health concern in Africa, where seasonal patterns are less defined and surveillance systems vary widely. Climatic factors such as temperature, humidity, and rainfall are known to influence influenza transmission dynamics, but their role across diverse African settings is not well understood. Moreover, the zoonotic nature of influenza, particularly the potential for spillover from animal reservoirs, adds complexity to its epidemiology, especially in regions with close human–animal interactions. Understanding how climate interacts with both human and zoonotic influenza dynamics is essential for strengthening early warning systems and guiding targeted interventions. This systematic review will synthesize existing evidence on the associations between climate-related parameters and influenza outcomes, including incidence, transmission potential, and severity. By clarifying these relationships, the review will support climate-informed public health planning and reinforce One Health approaches to influenza surveillance and control in Africa.

### Review objectives

1. Synthesize available evidence on associations between meteorological exposures (temperature, precipitation, humidity) and human influenza outcomes in Africa.
2. Compare reported associations across seasonal influenza subtypes and zoonotic influenza strains, noting gaps where evidence is absent.
3. Explore heterogeneity in findings by African subregion, climate zone, age group, surveillance platform, influenza subtype, and study design.

4. Assess the type and quality of evidence (quantitative vs. qualitative) and identify key research gaps and implications for public health planning, including integration of climate data into influenza surveillance and forecasting.

## Keywords

Influenza; Temperature; Precipitation; Humidity; Africa; Zoonotic influenza; Systematic review

## Country

Cameroon; Central African Republic; Côte d'Ivoire; Guinea; Madagascar; Niger; Senegal

## ELIGIBILITY CRITERIA

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### Population

#### *Included*

- Human participants of any age residing in African countries, in community or healthcare settings (outpatient, inpatient, sentinel surveillance, emergency departments).
- Studies reporting influenza outcomes (lab-confirmed or syndromic with virological confirmation in the same dataset or setting) linked to temperature, precipitation, and/or humidity.

#### *Excluded*

- Studies outside Africa or not disaggregated to an African location.
- Studies on other respiratory pathogens without influenza-specific outcomes.
- Purely laboratory bench studies without human or animal epidemiological data.

### Intervention(s) or exposure(s)

#### *Included*

This is an exposure-based review. The primary meteorological exposures include:

- Temperature: Daily mean, maximum, and minimum temperatures, apparent temperature, Heat and cold indices, temperature anomalies (e.g., deviations from climatological norms)
- Humidity: Relative humidity (%), Absolute humidity (g/m<sup>3</sup>), specific humidity, vapour pressure
- Precipitation: Daily rainfall (mm), Cumulative precipitation (weekly or monthly totals), Seasonal indicators (e.g., onset/duration of rainy or dry seasons)

### Comparator(s) or control(s)

#### *Included*

Where applicable, comparisons will be made between:

- Alternative exposure levels or categories (e.g., reference temperature, lowest decile of absolute humidity).
- Time periods with differing meteorological conditions (e.g., dry vs rainy season; heatwaves vs baseline).
- Spatial comparators across climate zones within Africa (when used by studies).
- Seasonal vs Zoonotic

### Study design

Only nonrandomized study types will be included.

#### *Included*

We would include the following types of studies:

- Observational epidemiologic designs: time-series, case-crossover, ecological panel studies, cohort studies (including birth cohorts), case-control, outbreak investigations with analytical assessment of weather correlates.
- Modelling studies that estimate quantitative exposure–response between the specified weather variables and human influenza outcomes based on observed data.
- Surveillance analyses (sentinel ILI/ARI/SARI with virologic confirmation) that model influenza activity against meteorological variables.

#### *Excluded*

We would exclude purely theoretical modelling without empirical data; editorials; non-analytic commentaries.

## Context

- Setting: All African regions (North, West, Central, East, Southern Africa), including island states (e.g., Madagascar, Mauritius, Seychelles, Cabo Verde, Comoros).
- Healthcare and community contexts: community surveillance, primary care, hospitals (general/tertiary), refugee/IDP camps where applicable.
- Climate diversity: arid, semi-arid, tropical rainforest/monsoon, subtropical highland, Mediterranean—capturing heterogeneity critical to seasonality.
- Exclusions: Non-African data; outcomes not attributable to influenza.

## SIMILAR REVIEWS

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### Check for similar records already in PROSPERO

*PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 14 January 2026). These are shown below along with the reasons given by that the review team for the reviews being different and/or proceeding.*

- The effects of environmental temperature and humidity on human blood pressure, blood sugar, and blood lipids: A meta-analysis [published 17 October 2025] [CRD420251158081]. The review was judged **not to be similar**
- Climate sensitivity of viral respiratory infections: a global systematic review and meta-analysis [published 19 October 2024] [CRD42024598878]. The review was judged **not to be similar**
- Associations Between Climate Variables and Melioidosis Incidence: A Systematic Risk Review of Environmental Risk Factors [published 5 May 2025] [CRD420251021930]. The review was judged **not to be similar**

## TIMELINE OF THE REVIEW

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### Date of first submission to PROSPERO

This record has not been submitted.

### Review timeline

Start date: 14 January 2026. End date: 14 May 2026.

### Date of registration in PROSPERO

This record has not been published.

## AVAILABILITY OF FULL PROTOCOL

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### Availability of full protocol

A full protocol has been written but is not available because:

*All aspects of the full protocol are already incorporated in the review registration questions*

## SEARCHING AND SCREENING

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### Search for unpublished studies

Only published studies will be sought.

### Main bibliographic databases that will be searched

The main databases to be searched are *Embase - Embase via Ovid, MEDLINE*.

*Other important or specialist databases that will be searched*

Web of Science Core collection; Global Health (CABI); African Index Medicus; African Journal Online

### Search language restrictions

There are no language restrictions.

### Search date restrictions

Databases will be searched for articles published from 1 January 1995 and before by 31 December 2025.

### Other methods of identifying studies

Other studies will be identified by: *contacting authors or experts*.

#### *Additional information about identifying studies*

We will hand-search reference lists of included papers and relevant reviews; and contact corresponding authors for missing data where necessary.

#### **Link to search strategy**

A full search strategy is not available.

#### **Selection process**

Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

#### **Other relevant information about searching and screening**

Screening and selection

- Rayyan will be used for dual-reviewer screening.
- Two reviewers will independently screen titles/abstracts and full texts against eligibility; disagreements will be resolved by consensus or a third reviewer.
- PRISMA 2020 flow diagram to document selection.

Data items to extract

- Bibliographic: author, year, journal, funding/conflicts.
- Study characteristics: country/region/city, setting (community, outpatient, inpatient), period, population, surveillance platform.
- Design & methods: study design, sample size, inclusion/exclusion, case definitions (ILI/ARI/SARI), lab methods (RT-PCR, antigen, culture), influenza type/subtype, zoonotic vs seasonal.
- Exposures: variable definitions (temperature, RH, AH, precipitation), units, sources (station/reanalysis), lag structure (DLNM, fixed lags), seasons/climate zones.
- Statistical approach: GLM/Poisson/Quasi-Poisson, GAM, DLNM, case-crossover, time-series decomposition, covariate adjustment (e.g., secular trends, holidays, school terms, population mobility, air pollution).
- Outcomes and effect estimates: measure type (RR/IRR/OR/% change), effect per unit exposure (and unit), CIs/SEs, p-values, model fit, non-linear parameters (spline knots), subgroup results.
- Risk of bias judgments (domain-level and overall).
- Notes: data gaps, correspondence with authors, conversion steps.

Data management

- Standardised extraction form piloted on 5–10 studies; forms stored in a shared repository.
- Unit harmonisation (e.g., temperatures in °C; Relative humidity in %; Absolute humidity in g/m<sup>3</sup>).

## **DATA COLLECTION PROCESS**

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#### **Data extraction from published articles and reports**

Data will be extracted independently by at least two people (or person/machine combination) with a process to resolve differences.

Authors will not be contacted for further information.

#### **Study risk of bias or quality assessment**

Risk of bias will be assessed using: *ROBINS-E*

Data will be assessed independently by at least two people (or person/machine combination) with a process to resolve differences.

Additional information will be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

### Reporting bias assessment

Each study will be rated as having Low, Moderate, or High risk of bias, with a brief rationale documented for each domain. Sensitivity analyses will be conducted to assess the influence of bias by excluding studies rated as High or Critical risk.

### Certainty assessment

The following effect measures will be used: Relative risks (RR), odds ratios (OR), incidence rate ratios (IRR), percent change per unit exposure (e.g., per 1 °C, per 10% RH, per 1 g/m<sup>3</sup> AH, per 10 mm precipitation), hazard ratios, or regression coefficients convertible to these metrics.

## OUTCOMES TO BE ANALYSED

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### Main outcomes

The key outcomes include influenza occurrence/transmission metrics associated with meteorological exposures, such as:

- Lab-confirmed influenza incidence or test positivity (%);
- ILI/ARI/SARI rates with virologic confirmation;
- Effective reproduction number or growth rates;
- Hospitalisation or mortality rates attributable to influenza

### Additional outcomes

- Seasonal characteristics: onset/peak timing, duration, amplitude of influenza seasons and their relation to weather.
- Subtype-specific differences: A(H1N1)pdm09 vs A(H3N2) vs B; zoonotic vs seasonal.
- Non-linear/threshold effects and interaction effects.
- Age-specific outcomes (e.g. below 5 years, school-age, adults, elderly).
- Healthcare burden proxies: outpatient visits, bed occupancy, ICU admissions attributable to influenza.

## PLANNED DATA SYNTHESIS

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### Strategy for data synthesis

A narrative (descriptive) synthesis will serve as the primary approach to summarizing findings from included studies. This synthesis will aim to identify and describe trends, associations, and patterns between climate-related parameters (e.g., temperature, precipitation, humidity) and human influenza outcomes (e.g., incidence, positivity rates, reproduction number, ILI/ARI/SARI rates, hospitalizations, deaths) across African contexts.

Where sufficient data are available, quantitative synthesis will be conducted via meta-analysis, using aggregate data extracted from eligible studies. Random-effects models will be applied to estimate pooled effect sizes (e.g., relative risks, odds ratios, incidence rate ratios) that describe associations between climatic exposures and influenza indicators. Heterogeneity will be assessed using the I<sup>2</sup> statistic. If enough studies are identified, subgroup analyses and meta-regression will be performed to explore sources of heterogeneity and examine the influence of study-level covariates (e.g., geographic setting, age group, influenza subtype, surveillance platform, study year).

To ensure robustness of the findings, sensitivity analyses will be conducted, including the exclusion of studies judged to be at high risk of bias.

## CURRENT REVIEW STAGE

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### Stage of the review at this submission

| Review stage                                        | Started | Completed |
|-----------------------------------------------------|---------|-----------|
| Pilot work                                          | ✓       |           |
| Formal searching/study identification               | ✓       |           |
| Screening search results against inclusion criteria |         |           |
| Data extraction or receipt of IPD                   |         |           |
| Risk of bias/quality assessment                     |         |           |

**Review stage****Started****Completed**

Data synthesis

**Review status**

The review is currently planned or ongoing.

**Publication of review results**

Results of the review will be published in English.

**REVIEW AFFILIATION, FUNDING AND PEER REVIEW**

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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### Review affiliation

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### Funding source

Review has no funding and no agreed support from an academic institution and is done in authors' own time.

### Peer review

There has been no peer review of this planned review.

## ADDITIONAL INFORMATION

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### Review conflict of interest

Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

### Medical Subject Headings

Africa; Humans; Influenza, Human; Seasons; Temperature; Rain; Humidity; Climate Change

### PROSPERO version history

No preview available

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